

Vivek Aruja

London, ON, CA | aruja.vivek@gmail.com | [linkedin.com/in/vivek-aruja](https://www.linkedin.com/in/vivek-aruja)

EDUCATION

University of Western Ontario

September 2023 – April 2028

Bachelor of Engineering Science in Electrical Engineering and AI Systems Engineering

London, ON

- Relevant Coursework: Power Systems, Electric Machines, Control Systems, Data Engineering and Machine Learning

TECHNICAL SKILLS

Programming: Python, MATLAB, Java, SQL

Power Systems: Load Flow & N-1 Contingency Analysis, Electrical Distribution Systems, Voltage Drop Analysis, System Optimization

Power & Grid Software: PowerWorld, OpenDSS, MicroStation, Tableau

Embedded & Hardware: ESP32, Fault Injection Testing, EVSE/EV Charging Diagnostics

Data & Reporting: Data Analysis, KPI Reporting, Microsoft Excel

EXPERIENCE

Hardware Test Engineer Intern

May 2026 – Present

Nanoleaf

Toronto, ON

- Improved connectivity reliability of an ESP32-based smart lighting prototype, as demonstrated through stable Wi-Fi reconnection and command handling, by implementing embedded control logic for addressable LED management and wireless interfaces.
- Enhanced system validation coverage and connectivity performance evaluation, as measured by regression test pass rates, by developing hardware diagnostic workflows using serial logging, fault injection analysis, and structured test matrices.
- Accelerated defect resolution and improved engineering data quality, as measured by structured bug tracking completeness, by documenting root causes, corrective actions, and retest results across hardware validation cycles.

Battery Team Member

May 2026 – Present

Western Sunstang Solar Car Team

London, ON

- Diagnosed OpenEVSE ground-fault errors on a J1772 charging interface through schematic to hardware wire tracing, continuity checks, chassis ground verification, and AC voltage measurements.
- Validated EVSE hardware connections including relay actuation, current sensing coil routing, chassis grounding, and proximity pilot wiring to isolate potential false ground fault detection causes.
- Documented troubleshooting findings and proposed next-step fixes, including firmware verification and current sensing coil inspection, to support safe EV charging hardware restoration.

Business Analyst

May 2025 – August 2025

Nanoleaf

Toronto, ON

- Increased overall sales by 15% and improved order processing efficiency, as measured by throughput and revenue KPIs, by developing and optimizing Shopify-based operational systems and process improvement initiatives.
- Improved real time coordination and operational reliability across stakeholders, as measured by reduction in manual communication overhead, by automating workflows using Shopify Flow and process optimization initiatives.
- Strengthened decision making and KPI reporting accuracy, as measured by a 22% increase in customer engagement, by analyzing large operational and sales datasets using predictive analytics and data driven methodologies.

PROJECTS

Transmission N-1 Contingency Analysis | *PowerWorld, Power Systems, Load Flow Analysis, Tableau, Excel/CSV*

- Built and solved a 13-bus transmission power flow model in PowerWorld, modeling generation sources, load centers, transformer corridors, and GTA/eastern transmission paths for N-1 reliability analysis.
- Evaluated 9 N-1 contingency cases by opening individual branches and tracking minimum bus voltage, maximum branch loading, severity classification, and corrective action requirements.
- Identified 5 severe contingencies, including a worst-case 0.645 pu voltage violation and 305.5% branch loading, then developed a Tableau dashboard and technical report to summarize system vulnerabilities.

MicroStation Distribution Service Connection Design Package | *MicroStation, Excel, Data Analysis*

- Produced a six sheet MicroStation drawing package for a 600V, 3-phase underground service connection, as measured by drawing completeness and revision tracking, by designing existing conditions, proposed route, plan/profile, single line, and construction detail sheets.
- Verified voltage drop and cable pull feasibility across an 85m duct bank route, as measured by a 1.28% voltage drop result, by developing Excel-based calculation workbooks for a 250 kVA underground secondary service.
- Improved design documentation accuracy and field readiness, as measured by project record and field verification checklist completeness, by cross checking drawing dimensions against calculations and preparing a final technical report.